

convenient (for numerical purposes) with respect to the overall structure. We will also discuss the transformation of a vector from the local coordinate system to the global coordinate system, using the concept of transformation matrices to express the stiffness matrix of an arbitrarily oriented bar element in terms of the global system. We will solve three example plane truss problems (see Figure 3–1 for a typical railroad trestle plane truss and a lift bridge truss over the Illinois River) to illustrate the procedure of establishing the total stiffness matrix and equations for solution of a structure.

Next we extend the stiffness method to include space trusses. We will develop the transformation matrix in three-dimensional space and analyze two space trusses. Then we describe the concept of symmetry and its use to reduce the size of a problem and facilitate its solution. We will use an example truss problem to illustrate the concept and then describe how to handle inclined, or skewed, supports.

We will then use the principle of minimum potential energy and apply it to rederive the bar element equations. We then compare a finite element solution to an exact solution for a bar subjected to a linear varying distributed load. We will introduce Galerkin's residual method and then apply it to derive the bar element equations. Finally, we will introduce other common residual methods—collocation, subdomain, and least squares—to merely expose you to them. We illustrate these methods by solving a problem of a bar subjected to a linear varying load.

3.1 Derivation of the Stiffness Matrix for a Bar Element in Local Coordinates

We will now consider the derivation of the stiffness matrix for the linear-elastic, constant cross-sectional area (prismatic) bar element shown in Figure 3–2. The derivation here will be directly applicable to the solution of pin-connected trusses. The bar is subjected to tensile forces T directed along the local axis of the bar and applied at nodes 1 and 2.

The bar element is assumed to have constant cross-sectional area A , modulus of elasticity E , and initial length L . The nodal degrees of freedom are local axial displacements (longitudinal displacements directed along the length of the bar) represented by u_1 and u_2 at the ends of the element as shown in Figure 3–2.

From Hooke's law [Eq. (3.1.1)] and the strain/displacement relationship [Eq. (3.1.2) or Eq. (1.4.1)], we write

$$\sigma_x = E\varepsilon_x \quad (3.1.1)$$

$$\varepsilon_x = \frac{du}{dx} \quad (3.1.2)$$

From force equilibrium, we have

$$A\sigma_x = T = \text{constant} \quad (3.1.3)$$

for a bar with loads applied only at the ends. (We will consider distributed loading in Section 3.10.) Using Eq. (3.1.2) in Eq. (3.1.1) and then Eq. (3.1.1) in Eq. (3.1.3) and

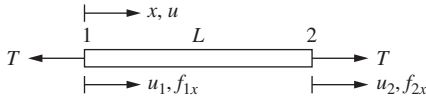


(a)



(b)

■ **Figure 3–1** (a) A typical railroad trestle plane truss (By Daryl L. Logan); (b) lift bridge truss over the Illinois River (By Daryl L. Logan)



■ **Figure 3–2** Bar subjected to tensile forces T ; positive nodal displacements and forces are all in the local x direction

differentiating with respect to x , we obtain the differential equation governing the linear-elastic bar behavior as

$$\frac{d}{dx} \left(AE \frac{du}{dx} \right) = 0 \quad (3.1.4)$$

where u is the axial displacement function along the element in the x direction and A and E are written as though they were functions of x in the general form of the differential equation, even though A and E will be assumed constant over the whole length of the bar in our derivations to follow.

The following assumptions are used in deriving the bar element stiffness matrix:

1. The bar cannot sustain shear force or bending moment, that is, $f_{1y} = 0$, $f_{2y} = 0$, $m_1 = 0$ and $m_2 = 0$.
2. Any effect of transverse displacement is ignored.
3. Hooke's law applies; that is, axial stress σ_x is related to axial strain ε_x by $\sigma_x = E\varepsilon_x$.
4. No intermediate applied loads.

The steps previously outlined in Chapter 1 are now used to derive the stiffness matrix for the bar element and then to illustrate a complete solution for a bar assemblage.

Step 1 Select the Element Type

Represent the bar by labeling nodes at each end and in general by labeling the element number (Figure 3–2).

As in deriving the spring element stiffness matrix, step 2 can be skipped at this time in deriving the one-dimensional bar element stiffness matrix. To facilitate the derivation, we can proceed directly to step 3

Step 3 Define the Strain/Displacement and Stress/Strain Relationships

The strain/displacement relationship is

$$\varepsilon_x = \frac{u_2 - u_1}{L} \quad (3.1.5)$$

and the stress/strain relationship by Hooke's law is

$$\sigma_x = E\varepsilon_x \quad (3.1.6)$$

Step 4 Derive the Element Stiffness Matrix and Equations

The element stiffness matrix is derived as follows. From elementary mechanics, we have

$$T = A\sigma_x \quad (3.1.7)$$

Now, using Eqs. (3.1.5) and (3.1.6) in Eq. (3.1.7), we obtain

$$T = AE \left(\frac{u_2 - u_1}{L} \right) \quad (3.1.8)$$

Also, by the nodal force sign convention of Figure 3–2,

$$f_{1x} = -T \quad (3.1.9)$$

When we substitute Eq. (3.1.8), Eq. (3.1.9) becomes

$$f_{1x} = \frac{-AE}{L}(u_2 - u_1) \quad (3.1.10)$$

Similarly,

$$f_{2x} = T \quad (3.1.11)$$

or, by Eq. (3.1.8), Eq. (3.1.11) becomes

$$f_{2x} = \frac{AE}{L}(u_2 - u_1) \quad (3.1.12)$$

Expressing Eqs. (3.1.10) and (3.1.12) together in matrix form, we have

$$\begin{Bmatrix} f_{1x} \\ f_{2x} \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} \quad (3.1.13)$$

Now, because $\{f\} = [k]\{d\}$, we have, from Eq. (3.1.13),

$$[k] = \frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad (3.1.14)$$

Equation (3.1.14) represents the stiffness matrix for a bar element in local coordinates. In Eq. (3.1.14), AE/L for a bar element is analogous to the spring constant k for a spring element.

Step 5 Assemble Element Equations to Obtain Global or Total Equations

Assemble the global stiffness and force matrices and global equations using the direct stiffness method described in Chapter 2 (see Section 3.6 for an example truss). This step applies for structures composed of more than one element such that (again)

$$[K] = \sum_{e=1}^N [k^{(e)}] \quad \text{and} \quad \{F\} = \sum_{e=1}^N \{f^{(e)}\} \quad (3.1.15)$$

where now all local element stiffness matrices $[k^{(e)}]$ must be transformed to global element stiffness matrices $[k]$ (unless the local axes coincide with the global axes) before the direct stiffness method is applied as indicated by Eq. (3.1.15). (This concept of coordinate and stiffness matrix transformations is described in Sections 3.3 and 3.4.)

Step 6 Solve for the Nodal Displacements

Determine the displacements by imposing boundary conditions and simultaneously solving a system of equations, $\{F\} = [K]\{d\}$.

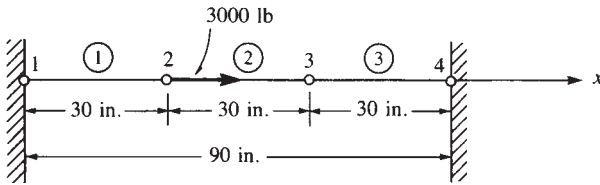
Step 7 Solve for the Element Forces

Finally, determine the strains and stresses in each element by back-substitution of the displacements into equations similar to Eqs. (3.1.5) and (3.1.6).

We will now illustrate a solution for a one-dimensional bar problem.

EXAMPLE 3.1

For the three-bar assemblage shown in Figure 3–3, determine (a) the global stiffness matrix, (b) the displacements of nodes 2 and 3, and (c) the reactions at nodes 1 and 4. A force of 3000 lb is applied in the x direction at node 2. The length of each element is 30 in. Let $E = 30 \times 10^6$ psi and $A = 1 \text{ in}^2$ for elements 1 and 2, and let $E = 15 \times 10^6$ psi and $A = 2 \text{ in}^2$ for element 3. Nodes 1 and 4 are fixed.



■ Figure 3–3 Three-bar assemblage

SOLUTION:

(a) Using Eq. (3.1.14), we find that the element stiffness matrices are

$$\begin{aligned}
 [k^{(1)}] &= [k^{(2)}] = \frac{(1)(30 \times 10^6)}{30} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 10^6 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \frac{\text{lb}}{\text{in.}} \\
 [k^{(3)}] &= \frac{(2)(15 \times 10^6)}{30} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 10^6 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \frac{\text{lb}}{\text{in.}}
 \end{aligned} \tag{3.1.16}$$

where, again, the numbers above the matrices in Eqs. (3.1.16) indicate the displacements associated with each matrix. Assembling the element stiffness matrices by the direct stiffness method, we obtain the global stiffness matrix as

$$[K] = 10^6 \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1+1 & -1 & 0 \\ 0 & -1 & 1+1 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \frac{\text{lb}}{\text{in.}} \tag{3.1.17}$$